

# Class12\_online

2026-02-14

## Section 1. Proportion of G/G in a population

Downloaded a CVS file from ENsembl < [https://useast.ensembl.org/Homo\\_sapiens/Variation/Sample?db=core;r=17:39835097-39955098;v=rs8067378;vdb=variation;vf=959672880#373531\\_tablePanel](https://useast.ensembl.org/Homo_sapiens/Variation/Sample?db=core;r=17:39835097-39955098;v=rs8067378;vdb=variation;vf=959672880#373531_tablePanel)

Here we may read this CVS file

```
mxl <- read.csv("373531-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
head(mxl)
```

```
## Sample..Male.Female.Unknown. Genotype..forward.strand. Population.s. Father
## 1 NA19648 (F) A|A ALL, AMR, MXL -
## 2 NA19649 (M) G|G ALL, AMR, MXL -
## 3 NA19651 (F) A|A ALL, AMR, MXL -
## 4 NA19652 (M) G|G ALL, AMR, MXL -
## 5 NA19654 (F) G|G ALL, AMR, MXL -
## 6 NA19655 (M) A|G ALL, AMR, MXL -
## Mother
## 1 -
## 2 -
## 3 -
## 4 -
## 5 -
## 6 -
```

```
table(mxl$Genotype..forward.strand.)
```

```
##
## A|A A|G G|A G|G
## 22 21 12 9
```

```
table(mxl$Genotype..forward.strand.) / nrow(mxl) * 100
```

```
##
## A|A A|G G|A G|G
## 34.3750 32.8125 18.7500 14.0625
```

Now let's look at a different population. I picked the GBR.

```
gbr <- read.csv("373522-SampleGenotypes-Homo_sapiens_Variation_Sample_rs8067378.csv")
```

Find proportion of G|G

```
round(table(gbr$Genotype..forward.strand.) / nrow(gbr) * 100, 2)
```

```
##  
##   A|A   A|G   G|A   G|G  
## 25.27 18.68 26.37 29.67
```

This variant that is associated with childhood asthma is more frequent in the GBR population than the MKL population. Let's now dig into this further.

## Section 4: Population Scale Analysis [HOMEWORK]

One sample is obviously not enough to know what is happening in a population. You are interested in assessing genetic differences on a population scale. So, you processed about ~230 samples and did the normalization on a genome level. Now, you want to find whether there is any association of the 4 asthma-associated SNPs (rs8067378...) on ORMDL3 expression.

How many samples do we have?

```
expr <- read.table("rs8067378_ENSG00000172057.6.txt")  
head(expr)
```

```
##   sample geno      exp  
## 1 HG00367  A/G 28.96038  
## 2 NA20768  A/G 20.24449  
## 3 HG00361  A/A 31.32628  
## 4 HG00135  A/A 34.11169  
## 5 NA18870  G/G 18.25141  
## 6 NA11993  A/A 32.89721
```

```
nrow(expr)
```

```
## [1] 462
```

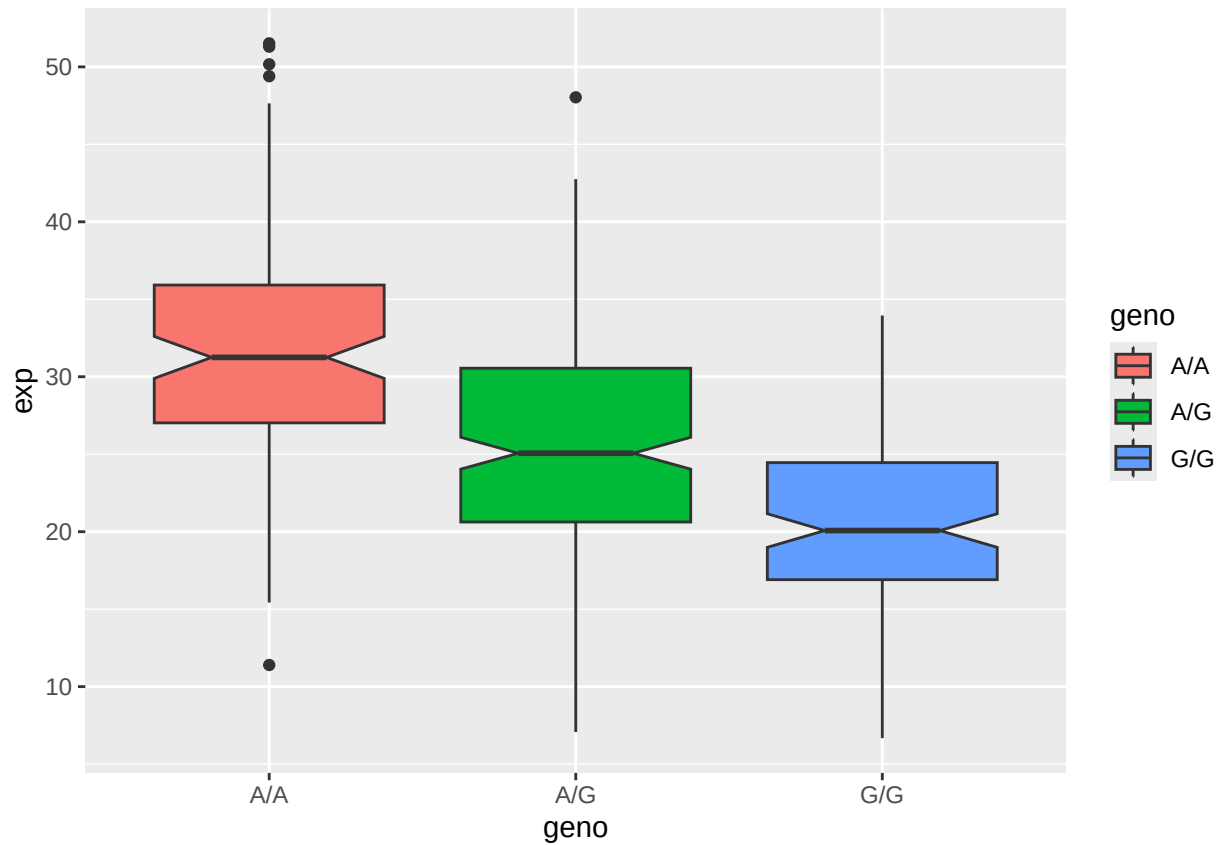
```
table(expr$geno)
```

```
##  
## A/A A/G G/G  
## 108 233 121
```

```
library(ggplot2)
```

Lets make a boxplot

```
ggplot(expr) + aes(x=geno, y=exp, fill=geno) +  
  geom_boxplot(notch=TRUE)
```



Q14: Generate a boxplot with a box per genotype, what could you infer from the relative expression value between A/A and G/G displayed in this plot? Does the SNP effect the expression of ORMDL3?

G/G expression is much lower than the expression of A/A, therefore, allele A could increase expression. Yes, the SNP does since there is no overlap in the medians.